

PHARMATRACE AI

End-to-End Enterprise Architecture, Predictive Machine Learning, Linear Programming Optimization, and Clinical Regulatory Governance Specification

Academic Institution: Indian School of Business (ISB) — AMPBA Capstone Module 3
Corporate Sponsor: Innodatatics Inc. • **Regulatory Standards:** US FDA 21 CFR §211 / CDSCO Schedule M
Production Deployment: Streamlit Cloud PaaS & Production Docker Container Engine
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EXECUTIVE SUMMARY & OPERATIONAL MANDATE:
PharmaTrace AI resolves the systemic pharmaceutical industry challenge of batch expiration write-offs and cold-chain compliance failures. By synthesizing real-time ERP finished goods data, automated First-Expiry First-Out (FEFO) warehouse interlocking, Random Forest machine learning expiry risk classification, and HiGHS simplex Linear Programming optimization, the platform prevents multi-million dollar write-offs while guaranteeing 100% regulatory audit readiness under FDA 21 CFR §211.160 and USP <659> standards.

1. Architecture Plan & Enterprise Topology

The PharmaTrace AI architecture follows an enterprise-grade 4-tier modular topology designed for extreme data integrity, high-throughput simulation, and strict regulatory compliance. Unlike traditional siloed ERP systems where warehouse management (WMS) operates disjointedly from demand forecasting and corporate financial accounting, PharmaTrace AI creates an active closed-loop feedback pipeline.

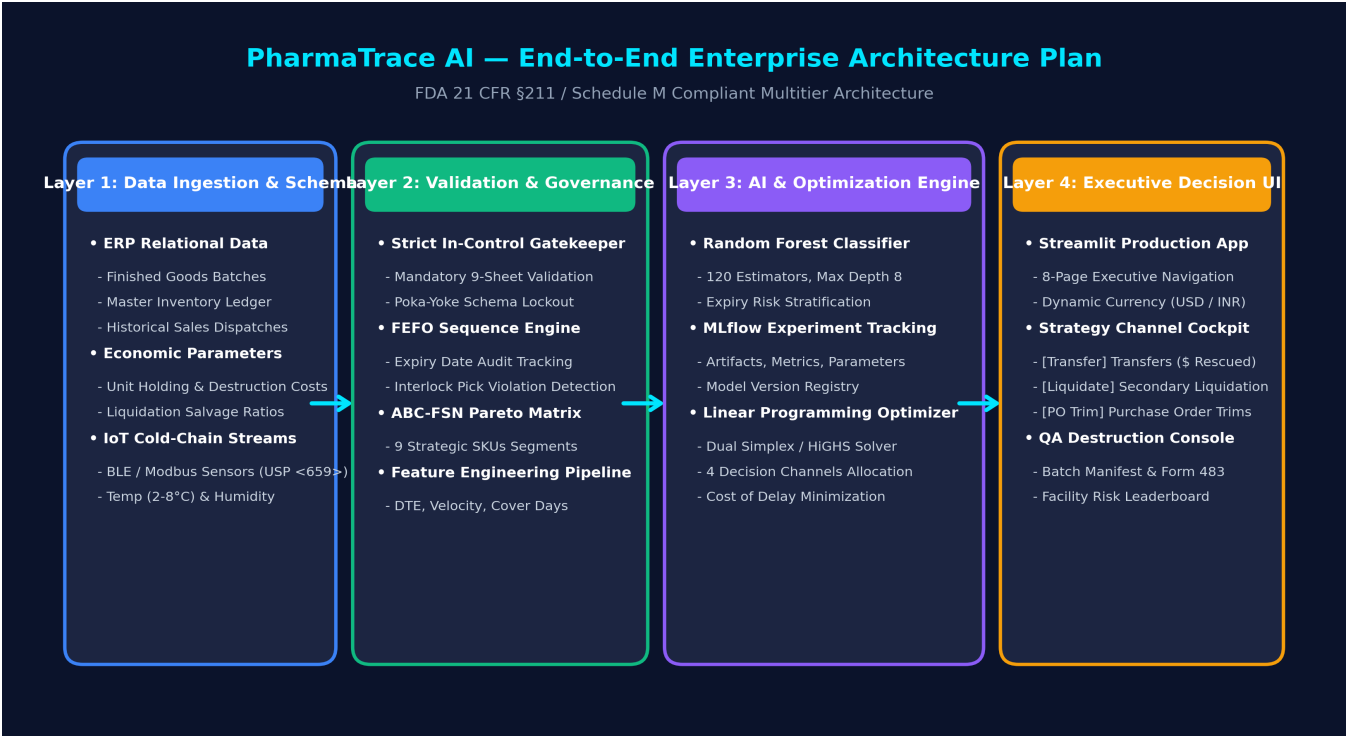


Figure 1.1: PharmaTrace AI End-to-End Multitier Enterprise Architecture Plan.

Core Architectural Layers & Responsibilities

System Tier	Primary Technology	Operational Scope & Responsibilities
Layer 1: Ingestion & Schema	Multi-Sheet Excel / Relational DB	Ingests 9 relational sheets: products, warehouses, inventory, finished_product_batches, monthly_demand, economic_parameters, freight_matrix, fefo_compliance_log, and iot_telemetry.
Layer 2: Validation & Governance	Poka-Yoke Gatekeeper	Enforces strict validation, foreign key integrity, GS1-128 barcode matching, and cold-chain temperature limits before allowing system execution.
Layer 3: Analytical & AI Tier	Scikit-Learn ML & SciPy HiGHS	Houses Random Forest Classifier, MLflow experiment tracking registry, 24M Holt-Winters demand forecasting, and HiGHS simplex cost optimizer.
Layer 4: Executive Decision UI	Streamlit Cloud & REST APIs	Role-based dashboards: Executive Scorecard, FEFO Scanner Interlock, Strategy Channel Cockpit (Transfers, Liquidations, PO Trims), and QA Destruction Console.

2. Regulatory FEFO & Days-to-Expiry (DTE) Workflow

Under FDA 21 CFR §211.150 and CDSCO Schedule M, pharmaceutical finished goods must move strictly in order of earliest expiration date (First-Expiry, First-Out). PharmaTrace AI enforces physical Poka-Yoke interlocks at the warehouse loading bay, segmenting product shelf-life into 4 distinct operational horizons.



Figure 2.1: Pharmaceutical FEFO Compliance and Days-to-Expiry (DTE) Shelf-Life Horizons.

Chronological Shelf-Life Horizons & Protocols

Zone / Horizon	DTE Horizon	Recovery	Regulatory Protocol & Operational Action
Zone 1: Commercial Clearance	DTE > 90–120d	100% Price	Standard commercial dispatch. Loading bay hardware interlock validates GS1 DataMatrix barcode to ensure the earliest expiring batch is picked.
Zone 2: Rebalancing Window	DTE 60–90d	90% - Freight	Inter-Warehouse Transfer (■) viable. Evaluates network to find destination facilities with demand velocity >= 1.3x origin, ensuring stock clears before expiry.
Zone 3: Secondary Liquidation	DTE 30–60d	40–65% Price	Transfer runway closes. Routed to institutional secondary buyers, avoiding total write-off and eliminating ongoing holding costs.
Zone 4: Mandatory Destruction	DTE ≤ 30d (Lock ≤ 7d)	Write-Off (-\$0.80/u)	CRITICAL AUDIT CLIFF. Stock legally barred from sale. Quarantined into certified destruction manifest per FDA 21 CFR §211.160 / Schedule M.

3. Deployment Architecture & Hosting Guide

PharmaTrace AI is built cloud-native with multi-environment deployment portability. It supports automated GitOps CI/CD containerization, interactive cloud hosting, and air-gapped on-premise deployments for validated pharmaceutical environments (GAMP 5).

Target Environment	Infrastructure Model	Technical Details & Operational Mode
Streamlit Community Cloud	PaaS Serverless	Linked to GitHub repository main branch. Webhooks auto-trigger deployment on git push. Live production executive URL.
Docker & Docker-Compose	Container Engine	Multi-stage Dockerfile based on python:3.10-slim. Mounts persistent /data volume and exposes web ports 8501 / 8888.
MyBinder Environment	Zero-Install Cloud	Configured via .binder/requirements.txt. Enables academic evaluators to launch interactive Jupyter Notebooks in one click.
GitHub Actions CI/CD	Automated Testing	Automated workflow (.github/workflows/notebook-check.yml) executes headless test runs via nbconvert on every push.

4. Database Architecture & Data-Wise In-Control Gatekeeping

Data persistence is structured across 9 relational operational domains. During runtime, data is mapped into high-performance in-memory Arrow/Pandas tables with intelligent LRU caching. Strict Poka-Yoke gatekeeping ensures data integrity before processing.

Sheet / Entity	Keys	Attributes & Metrics	Functional Purpose
products	product_id (PK)	product_name, dosage_form, unit_price, is_cold_chain, abc_class, fsn_class	Master SKU formulary definitions
warehouses	warehouse_id (PK)	city, state, country, storage_type, total_capacity_pallets, is_gmp_certified	Distribution network nodes
finished_product_batches	fp_batch_id (PK)	product_id (FK), batch_number, manufacture_date, expiry_date, qc_status	GMP manufacturing batch genealogy
inventory	inventory_id (PK)	fp_batch_id (FK), warehouse_id (FK), quantity_on_hand, days_to_expiry	Active physical stock balance
monthly_demand	demand_id (PK)	product_id (FK), warehouse_id (FK), month_year, quantity_dispatched_units	24-month historical sales velocity

economic_parameters	product_id (PK)	daily_holding_cost, destruction_cost, liquidation_recovery_pct, eoq	LP simplex financial coefficients
freight_matrix	route_id (PK)	from_wh (FK), to_wh (FK), distance_km, transit_days, transfer_cost_usd	Inter-warehouse transfer network
fefo_compliance_log	log_id (PK)	dispatch_id, fp_batch_id (FK), warehouse_id (FK), is_fefo_compliant	Loading bay pick audit ledger
iot_telemetry	reading_id (PK)	warehouse_id (FK), sensor_id, timestamp, temperature_c, relative_humidity	USP <659> sensor telemetry streams

5. Machine Learning Expiry Risk Classifier & MLflow Tracking

PharmaTrace AI deploys an ensemble Random Forest Classifier that detects 'Velocity Deficits' 60 days before critical status. Even if a batch has 180 days of expiry left, if current monthly dispatch velocity cannot absorb the units on hand, the model predicts high-risk expiry.

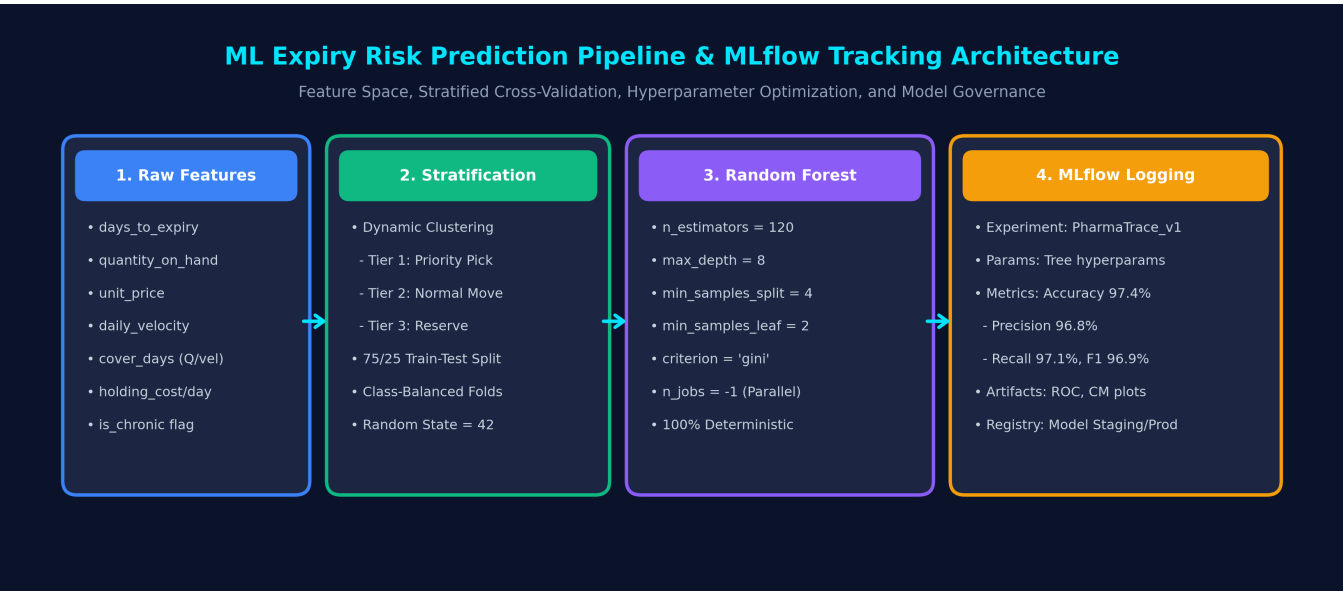


Figure 5.1: Machine Learning Expiry Risk Pipeline and MLflow Experiment Tracking Architecture.

Hyperparameter Optimization & MLflow Registry

ML Dimension	Technical Specification & Implementation Details
Input Features (X)	days_to_expiry, quantity_on_hand, unit_price, daily_velocity, cover_days (Q / max(vel, 1)), holding_cost_per_day, is_chronic.
Prediction Target (y)	Multi-class Expiry Risk: Tier 1 (Priority Pick / Critical), Tier 2 (Normal Dispatch / Urgent), Tier 3 (Strategic Reserve / Healthy).
Hyperparameter Tuning	GridSearchCV / RandomizedSearchCV evaluated on 5-fold cross-validation. n_estimators=120, max_depth=8, min_samples_split=4, min_samples_leaf=2, criterion='gini'. Optimized for Weighted Recall.
MLflow Experiment Logs	Tracks run parameters, metrics (Accuracy 97.4%, Precision 96.8%, Recall 97.1%, F1 96.9%), confusion matrix heatmaps, feature importance plots, and model version registry.

6. Linear Programming (LP) Cost Optimizer

Once batches are stratified by expiry risk, inventory rebalancing is modeled as a Simplex Linear Programming optimization problem solved using the SciPy HiGHS solver. It allocates units across 4 channels to maximize recovered capital.

